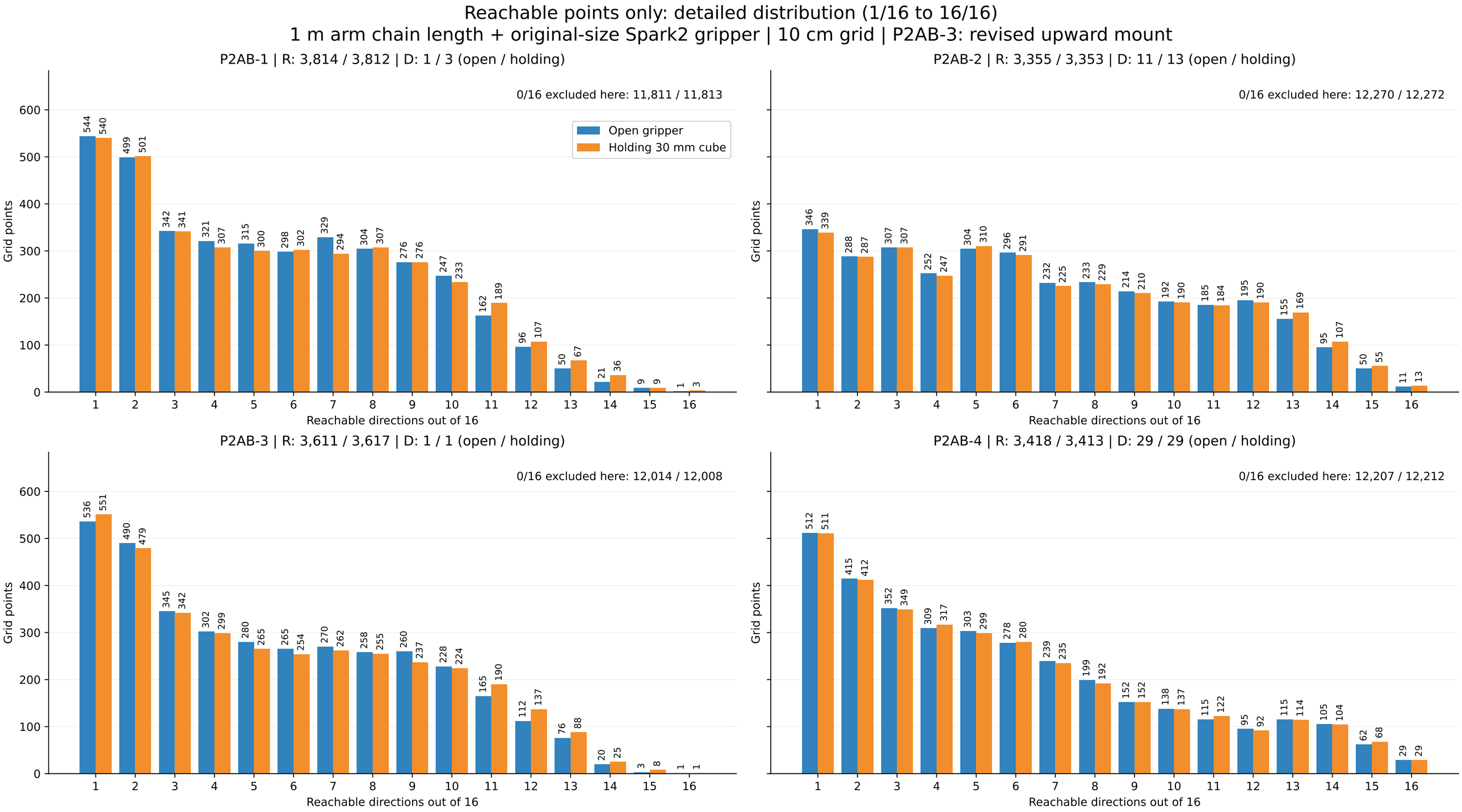
Reachable points only: detailed distribution (1/16 to 16/16)
1 m arm chain length + original-size Spark2 gripper | 10 cm grid | P2AB-3: revised upward mount
P2AB-1 | R: 3,814 / 3,812 | D: 1 / 3 (open / holding)
0/16 excluded here: 11,811 / 11,813
Open gripper
Holding 30 mm cube
Grid points
Reachable directions out of 16
544 540
499 501
342 341
321 307
315 300
298 302
329 294
304 307
276 276
247 233
162 189
96 107
50 67
21 36
9 9
1 3
P2AB-2 | R: 3,355 / 3,353 | D: 11 / 13 (open / holding)
0/16 excluded here: 12,270 / 12,272
346 339
288 287
307 307
252 247
304 310
296 291
232 225
233 229
214 210
192 190
185 184
195 190
155 169
95 107
50 55
11 13
P2AB-3 | R: 3,611 / 3,617 | D: 1 / 1 (open / holding)
0/16 excluded here: 12,014 / 12,008
536 551
490 479
345 342
302 299
280 265
265 254
270 262
258 255
260 237
228 224
165 190
112 137
76 88
20 25
3 8
1 1
P2AB-4 | R: 3,418 / 3,413 | D: 29 / 29 (open / holding)
0/16 excluded here: 12,207 / 12,212
512 511
415 412
352 349
309 317
303 299
278 280
239 235
199 192
152 152
138 137
115 122
95 92
115 114
105 104
62 68
29 29